



Solving linear equations graphically using technology

Task A: Solving equations with unknowns one side

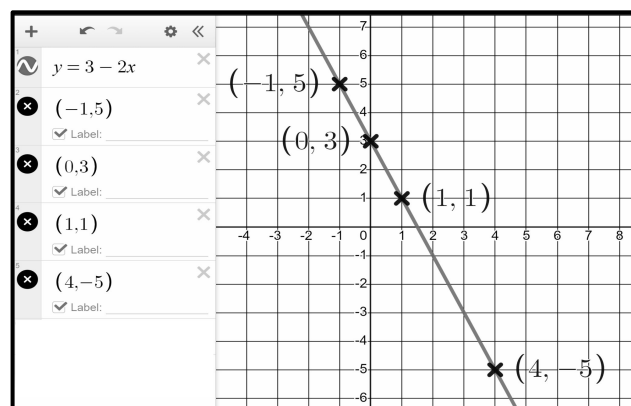
1) Draw the line $y = 3 - 2x$ and use it to solve the below equations.

a) $3 - 2x = 1$ $x = 1$

b) $3 - 2x = -5$ $x = 4$

c) $3 - 2x = 5$ $x = -1$

d) $3 - 2x = 3$ $x = 0$

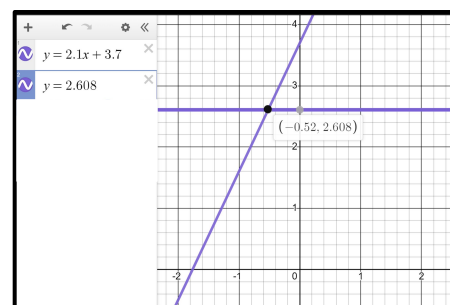
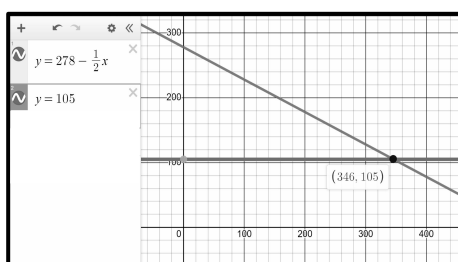
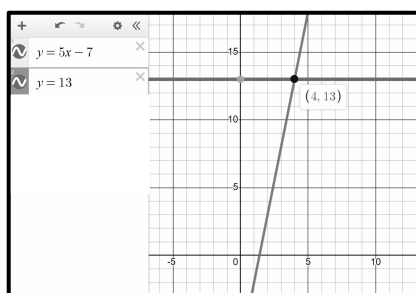


2) Draw two lines for each to find the solutions to these equations:

a) $5x - 7 = 13$
 $x = 4$

b) $278 - \frac{1}{2}x = 105$
 $x = 346$

c) $2.1x + 3.7 = 2.608$
 $x = -0.52$

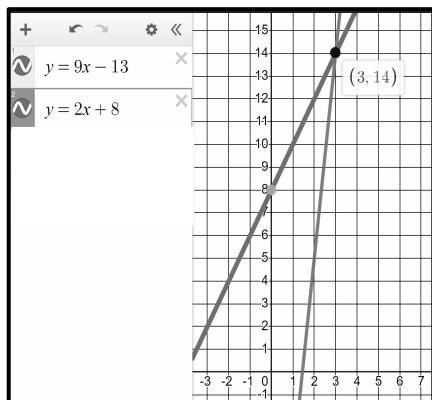


Task B: Solving equations with unknowns on both sides

1) Plot pairs of lines and find the intersections to solve the below equations:

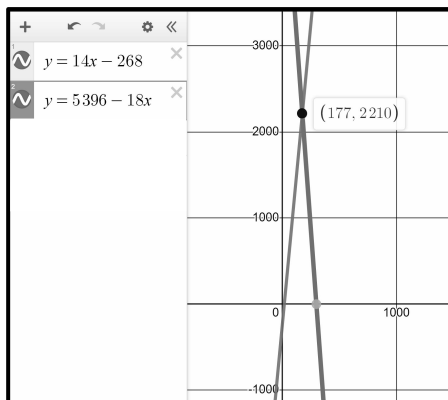
a) $9x - 13 = 2x + 8$

Intersect at (3, 14) therefore
solution is $x = 3$



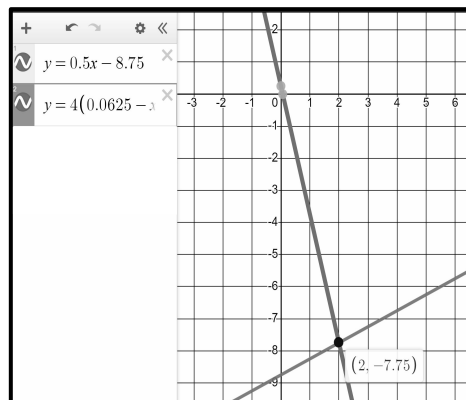
b) $14x - 268 = 5396 - 18x$

Intersect at (177, 2210)
therefore solution is $x = 177$



c) $0.5x - 8.75 = 4(0.0625 - x)$

Intersect at (2, -7.75)
therefore solution is $x = 2$



2) Alex says this graph shows that
the solution to the equation
 $3x + 9 = -2x - 1$ is $x = 3$

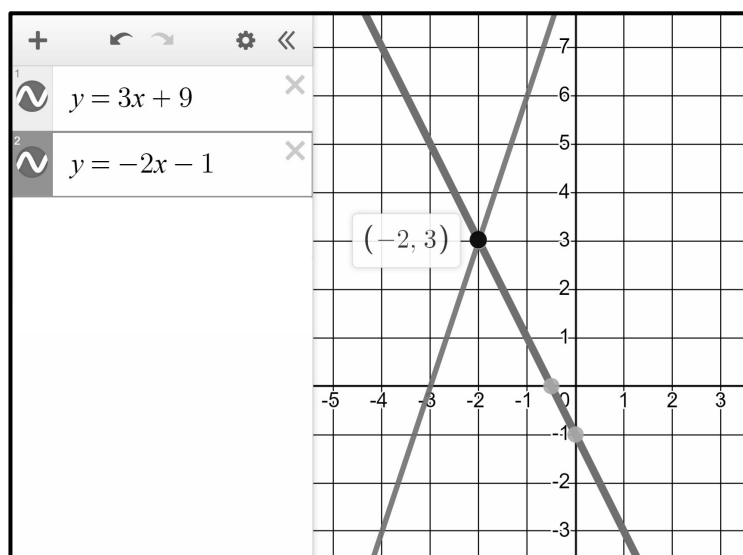


Alex is wrong.

Identify the correct solution and write a
sentence explaining Alex's error to him.

You might have written:

- $x = -2$ is the solution
- 3 is the value of both expressions
when $x = -2$



3) One of the below equations has no solution.

Identify which one it is and show why it has no solution.

a) $8 - 3x = 3x + 2$

Different gradients therefore intersection and a
solution.

b) $11x - 25 = 25x + 7$

Different gradients therefore intersection and a
solution.

c) $6x - 11 = 2(3x + 4)$

$6x - 11 = 6x + 8$ Same gradient. No intersection, no solution.

